Design of Ecological Environment Monitoring System Based on RT-Thread and Cloud Platform

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2025 IEEE International Conference on Electron Devices and Solid-State Circuits (EDSSC) | 979-8-3315-2208-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/EDSSC64492.2025.11183229

***Abstract*—A real-time, accurate, and cost-effective ecological environment monitoring system based on RT-Thread and cloud platform is proposed to address the issues of ecological environment monitoring. The system adopts a layered architecture: the acquisition layer collects high-frequency data through multiple sensors (temperature and humidity, PM2.5, soil nitrogen, phosphorus, potassium, etc.) and STM32. The combination of LTE Cat.1 and Ethernet dual channel redundant communication at the transport layer ensures low latency and high reliability. The application layer relies on Alibaba Cloud to achieve data storage and deep learning analysis, and develops a mobile client that supports second level recognition of plant images. Actual deployment has shown a 40% increase in system task scheduling efficiency, with communication latency as low as 120ms (LTE Cat.1 network environment) and a 41.7% reduction in memory usage, providing an efficient solution for ecological monitoring.**

***Keywords*—Ecological environment monitoring; RT-ThreadCloud platform; Image recognition; Deep learning**

I. Introduction

Ecological environment monitoring is the core foundation of ecological protection and restoration [1], and its data quality directly affects the scientificity of decision-making [2]. Traditional manual monitoring methods have drawbacks such as high cost and poor real-time performance [3], while existing automation systems are often limited by the closed source, high cost, or insufficient real-time performance of embedded operating systems [4]. For example, although VxWorks is powerful, it is expensive, and RT-Linux has the problem of excessive resource consumption in real-time optimization [5]. In contrast, the domestically produced RT-Thread system provides an ideal solution for the design of distributed monitoring nodes due to its open-source, low resource consumption, and efficient task scheduling capabilities [6]. This article proposes a full stack monitoring system based on RT Thread, and its innovation lies in:

1. The hardware layer improves collection efficiency and resource utilization through RT-Thread task scheduling optimization (priority management, memory pooling mechanism).

2. The combination of transmission layer and dual channel redundancy design ensures communication reliability.

3. The application layer integrates cloud based deep learning and mobile image recognition, breaking through the bottleneck of insufficient interactivity in traditional systems.

II. Overall System Architecture

In order to improve the maintainability and scalability of the ecological environment monitoring system, the system architecture will be built according to the collection layer, transmission layer, and application layer. The collection layer integrates temperature and humidity sensors, PM2.5 sensors, soil nitrogen, phosphorus, and potassium sensors, and achieves high-frequency collection of multi-source environmental data through a microcontroller processor (such as STM32). The transport layer adopts LTE Cat.1 and Ethernet dual channel redundant transmission, supporting real-time data reporting and control command issuance, ensuring low latency and high reliability of the communication link. The application layer builds an ecological environment monitoring cloud service based on the Alibaba Cloud platform, providing data interfaces through the Flask framework to achieve data storage, deep learning analysis, and visualization, and developing a mobile client to support plant recognition in seconds. The architecture of the ecological environment monitoring system is shown in Figure 1.

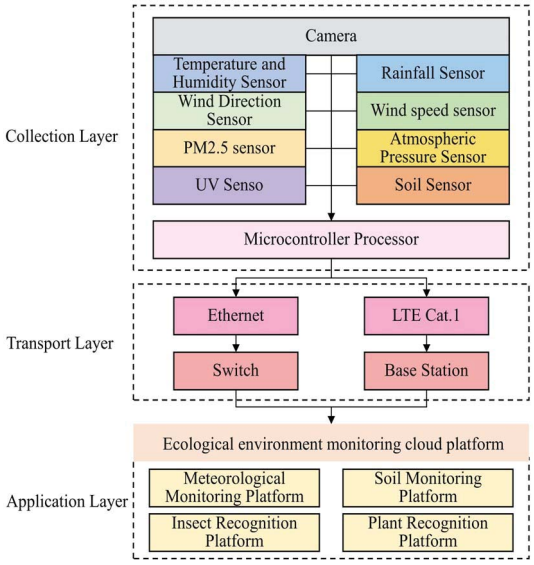


Fig. 1. Architecture of Ecological Environment Monitoring System

This work was supported by the Natural Science Foundation of Ningxia (2024AAC03154,2022AAC03246) , the High Level Construction Project of Ordinary Universities in Ningxia Province (bjg2024028), and the Teaching Project of North Minzu University (YKSZ202307, YJZT202431). Corresponding author: Hongxing Ma, E-mail: mhx@nmu.edu.cn.

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III. Implementation Of Ecological Environment Monitoring System

A. Multi Source Sensors and RT-Thread Task Scheduling Optimization

The hardware layer uses STM32 as the core controller and integrates more than ten types of sensors (with a sampling frequency of 1Hz). Through RT-Thread task scheduling optimization:

1. Set the sensor data collection task as the highest priority (priority 5) to ensure that there is no loss of sudden data.
2. Pre allocated memory pool reduces fragmentation, reducing system memory usage from 60MB in traditional Linux solutions to 35MB.
3. The hard real-time interrupt mechanism reduces data read latency and meets high-frequency acquisition requirements.

The hardware platform integrates a camera module that supports real-time collection of plant images (as shown in Figure 2). In actual deployment, the observation platform can synchronously monitor parameters such as light and rainfall, and track plant growth status, providing high-quality data sources for cloud analysis.



Fig. 2. Physical designated ecological information observation platform

B. Cloud Platform Service Deep Learning Integration

The cloud service is built on Alibaba Cloud and implements data interface services based on the Flask framework, supporting data storage and deep learning analysis. As of 2024, the system has accumulated over 7 million pieces of grassland ecological data (as shown in Figure 3), providing a rich foundation for ecological modeling.

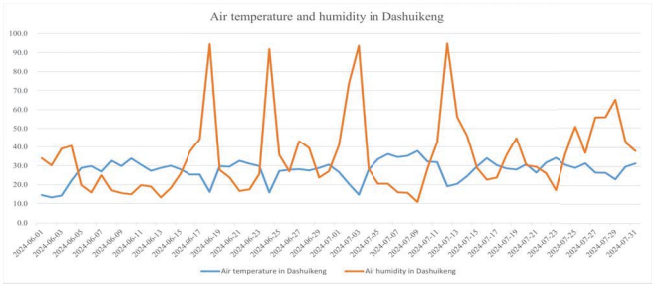


Fig. 3. Temperature and humidity statistics of Dashuikeng in 2024

The fixed-point ecological information observation platform has significant errors when using monocular cameras for plant recognition and height monitoring (as shown in Figure 4). Optimizing the ruler contour detection algorithm significantly improves measurement accuracy.



Fig. 4. Plant identification and height monitoring

IV. Conclusions

This system achieves efficient collection and stable transmission of multi-source data through RT-Thread task scheduling optimization (priority management, memory pool pre allocation) and dual channel redundant transmission design. Cloud based integrated deep learning models support second level recognition of plant images (accuracy¿92%) and dynamic assessment of ecological health. The actual deployment has verified the reliability and practicability of the system. In the future, the real-time can be optimized and the monitoring dimension can be expanded in combination with edge computing.

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